

Final Project: Individual Case Safety Report (ICSR)

Pharmacovigilance Assessment — VirtualWorks PV Internship

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Brief: As a Drug Safety Associate, a complete safety evaluation has been performed on the case below, consolidating ADR identification, ICSR reporting, causality assessment, and seriousness/expectedness classification.

Case Scenario

Patient: S.K., 45-year-old female, 62 kg. **Indication:** Community-acquired respiratory tract infection (RTI). **Suspected drug:** Amoxicillin + Clavulanic Acid (Augmentin) 625 mg, oral, twice daily, started 23/06/2026.

On Day 3 of therapy (26/06/2026), the patient developed a diffuse erythematous maculopapular rash over the trunk and upper limbs, mild facial edema, and generalized pruritus. The suspected drug was discontinued immediately. She was treated symptomatically with Cetirizine 10 mg oral and topical corticosteroid cream, with complete resolution within 5 days (01/07/2026). No hospitalization was required. Relevant history: known childhood hypersensitivity to sulfonamides; no prior documented penicillin allergy; no pregnancy; no hepatic/renal dysfunction; non-smoker, no alcohol use.

1. ADR Identification

Parameter	Detail
Reaction observed	Diffuse erythematous maculopapular rash, mild facial edema, generalized pruritus
Onset	72 hours (Day 3) after starting Amoxicillin-Clavulanate 625 mg BD
Suspected drug	Amoxicillin + Clavulanic Acid (Augmentin) 625 mg
Reaction type	Type I/IV hypersensitivity (drug-induced exanthem with angioedema feature)
Action taken	Drug discontinued on Day 3
Treatment given	Cetirizine 10 mg oral + topical corticosteroid cream
Outcome	Recovered (complete resolution in 5 days; no hospitalization)

Clinical note: The combination of cutaneous rash with facial edema raises clinical suspicion of an early angioedema component, even though it resolved without airway compromise. This was specifically monitored and documented, which is significant for the seriousness classification in Section 4.

2. Individual Case Safety Report (ICSR)

A. Patient Information

Patient Initials	S.K.
Age / Sex	45 years / Female
Weight	62 kg
Relevant medical history	Known childhood hypersensitivity to sulfonamides; no penicillin allergy on record; not pregnant; no hepatic/renal dysfunction; non-smoker; no alcohol use

B. Suspected Medication

Drug (Brand/Generic)	Augmentin (Amoxicillin + Clavulanic Acid)
Manufacturer / Batch	GSK / AUG-2024-B77 (Exp. 09/2026)
Dose, Route, Frequency	625 mg, Oral, Twice daily (BD)
Therapy dates	Started 23/06/2026 – Stopped 26/06/2026
Reason for use	Community-acquired respiratory tract infection
Concomitant medication	None (Cetirizine and topical corticosteroid were used to treat the reaction, not concomitant therapy)

C. Reaction & Outcome

Date of onset	26/06/2026 (Day 3 of therapy)
Date of recovery	01/07/2026
Dechallenge	Positive – reaction abated after drug was stopped
Rechallenge	Not performed – clinically inappropriate given hypersensitivity reaction
Laboratory data	None performed; diagnosis was clinical, confirmed by treating physician on 26/06/2026
Outcome	Recovered

D. Reporter Details

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3. Causality Assessment

Two standard, internationally recognized causality assessment methods were applied: the **Naranjo Algorithm** (score-based, widely used by clinical researchers/CROs) and the **WHO-UMC Scale** (criteria-based, used by regulatory agencies including CDSCO).

Naranjo Algorithm

#	Question	Score
1	Previous conclusive reports of this reaction?	+1
2	ADR appeared after the suspected drug was administered?	+2
3	ADR improved when drug was discontinued (dechallenge)?	+1

4	ADR reappeared on rechallenge? (Not performed – clinically inappropriate)	0
5	Alternative causes ruled out?	+1
6	ADR confirmed by placebo? (N/A)	0
7	Drug detected in toxic concentration in blood/fluids?	0
8	Dose-response relationship demonstrated?	0
9	Prior similar reaction to same/related drug? (Sulfonamide cross-reactivity history)	+1
10	Confirmed by objective evidence? (Physician-confirmed rash on exam)	+1
TOTAL		+7

Naranjo Classification: PROBABLE (Score range 5–8)

WHO-UMC Scale

WHO-UMC Criteria	Met?
Plausible time relationship (Day 3 – consistent with beta-lactam hypersensitivity latency)	Yes
Known reaction type for the drug (listed in Augmentin SmPC)	Yes
Reaction improved on dechallenge (resolved in 5 days)	Yes
No plausible alternative cause	Yes
Rechallenge confirmed (not performed – ethically inappropriate)	No

WHO-UMC Classification: PROBABLE – "Certain" requires rechallenge confirmation, which is ethically inappropriate in an allergic/hypersensitivity reaction. Probable is the highest achievable classification in this spontaneous report.

4. Seriousness & Expectedness Classification

Parameter	Classification	Justification
Seriousness	Serious	Although the patient did not require hospitalization, mild facial edema raised clinical concern for ea
Expectedness	Expected	Cutaneous hypersensitivity reactions (rash, pruritus, angioedema) are well-documented, labeled a

Regulatory reporting implication: As a Serious, Expected ADR, this case requires reporting within standard expedited timelines (15 calendar days) to CDSCO/regulatory authority, with the ICSR and causality assessment documented above as supporting evidence.

Conclusion

The reaction (maculopapular rash with facial edema following Amoxicillin-Clavulanate) was identified, fully documented via ICSR, and assessed using two independent causality methods, both converging on a **Probable** causal association. The event has been classified as **Serious** (medically important) and **Expected** (labeled reaction), and is recommended for expedited regulatory submission. Accurate reporting today contributes directly to safer patient outcomes tomorrow.